

Review Sentiment Toward AI Coding Agents

Section draft prepared by Amrinder Singh | AIDev / AIDev-pop dataset

Purpose

This section contributes the review-sentiment part of the team's broader analysis of AI coding agents in open-source pull requests. It should be combined with the team's results on merge success, review duration, task category, PR complexity, repository context, and developer experience.

Research Question

How does review sentiment toward AI coding agents differ across agents, and how has that sentiment evolved over time in open-source pull request discussions?

Sub-questions:

- How has review sentiment toward AI coding agents changed over time?
- Which AI coding agents receive the most positive and most negative review sentiment in pull request reviews and comments?
- To what extent do different sentiment-analysis methods affect the observed sentiment trends and agent rankings?

Data and Method

The analysis uses AIDev-pop, which contains 33,596 agent-authored pull requests from repositories with more than 100 stars. The relevant tables were `pull_request.parquet`, `pr_comments.parquet`, `pr_reviews.parquet`, and `pr_review_comments_v2.parquet`.

The analysis focuses on human external review discussion. Bot comments, PR-author self-comments, empty comments, and very short comments were removed. Markdown code blocks, inline code, URLs, and generated markup were cleaned before sentiment analysis.

Final filtered sample: **21,922 human external review/comment text items**.

Source	Count
Inline review comments	10,830
PR issue-thread comments	8,900
Top-level review bodies	2,192

Agent	Text items
GitHub Copilot	13,957
Devin	5,194
OpenAI Codex	1,720
Cursor	617
Claude Code	434

Four sentiment methods were compared:

Method	What it captures
VADER	General text sentiment
TextBlob	General text polarity
SentiCR-trained classifier	Code-review-specific negative vs non-negative sentiment
Review-state proxy	Formal GitHub review outcome: approved, changes requested, or commented

Key Results

Sentiment Over Time

Review sentiment is generally neutral-to-positive, but it becomes less strongly positive over time. The earliest month is very positive but has only 41 comments, so it should be interpreted cautiously.

Month	N	VADER mean	TextBlob mean	SentiCR negative share
2024-12	41	0.665	0.160	9.8%
2025-01	517	0.104	0.059	24.6%
2025-02	538	0.200	0.074	24.5%
2025-03	742	0.176	0.075	22.6%
2025-04	975	0.141	0.068	20.7%
2025-05	4,086	0.090	0.055	25.6%
2025-06	7,322	0.099	0.054	24.8%
2025-07	7,570	0.095	0.058	21.9%
2025-08	131	0.002	0.070	15.3%

Agent-Level Sentiment

General text sentiment ranks Claude Code highest and GitHub Copilot lowest.

Agent	VADER mean	Positive share	Negative share
Claude Code	0.269	57.8%	13.4%
Cursor	0.147	48.5%	22.0%
Devin	0.141	39.0%	16.7%
OpenAI Codex	0.119	49.8%	23.3%
GitHub Copilot	0.080	36.6%	20.6%

Code-review-specific sentiment and formal review states instead favor OpenAI Codex and Cursor.

Agent	SentiCR negative share	Review-state score	Approved	Changes requested
OpenAI Codex	17.8%	0.613	66.5%	5.3%

Cursor	20.1%	0.592	63.3%	4.1%
Devin	22.3%	0.334	40.3%	6.9%
Claude Code	24.7%	0.366	41.1%	4.5%
GitHub Copilot	24.9%	0.239	36.0%	12.1%

Method Sensitivity

The answer changes depending on the sentiment method.

Method	Ranking from most positive to least positive
VADER	Claude Code > Cursor > Devin > OpenAI Codex > Copilot
TextBlob	Claude Code > Cursor > OpenAI Codex > Devin > Copilot
SentiCR-trained classifier	OpenAI Codex > Cursor > Devin > Claude Code > Copilot
Review-state proxy	OpenAI Codex > Cursor > Claude Code > Devin > Copilot

This means the report should avoid claiming that one agent is simply "most trusted." General language sentiment and code-review-specific sentiment capture different parts of reviewer response.

Statistical and Robustness Evidence

Sentiment differences across agents were statistically significant:

Test	Result
Kruskal-Wallis on VADER scores	H = 178.657, p = 1.449e-37
Kruskal-Wallis on TextBlob scores	H = 132.305, p = 1.252e-27

Method agreement was limited:

Method pair	Spearman correlation
VADER vs TextBlob	0.441
VADER vs SentiCR-trained	0.075
TextBlob vs SentiCR-trained	0.063

Dataset imbalance affects the strength of the rankings. Raw and equal-agent downsampled results were stable, but repository-normalized and within-repository analyses changed some rankings. The within-repository subset was much smaller: 1,841 comments from 19 repositories.

Bootstrap checks strengthened the method-specific conclusions:

Method	Stable bootstrap finding
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VADER	Claude Code ranked 1st in 100% of samples; Copilot ranked last in 100%
TextBlob	Claude Code ranked 1st in 93.8% of samples; Copilot ranked last in 100%
SentiCR-trained	OpenAI Codex ranked 1st in 89.9% of samples

Link to Merge Outcome

Merged PRs had lower code-review-specific negativity than unmerged PRs.

PR outcome	PRs	VADER mean	TextBlob mean	SentiCR negative share
Not merged	2,746	0.137	0.056	22.4%
Merged	3,225	0.144	0.085	18.1%

In a logistic regression predicting merge outcome, SentiCR negative share was significantly associated with lower merge odds, coefficient = -0.479, $p = 4.062e-08$. VADER was not significant, $p = 0.297$. This suggests that code-review-specific negativity is more closely linked to PR outcomes than general positive wording.

Final Section Conclusion

Review sentiment toward AI coding agents is generally neutral-to-positive, but it differs by agent, sentiment method, and repository context. General text sentiment gives a stable result that Claude Code receives the warmest review tone and GitHub Copilot receives the lowest general tone.

Code-review-specific and formal review signals instead favor OpenAI Codex and Cursor, suggesting that these agents receive less technically negative review feedback or stronger approval outcomes.

The strongest conclusion is not that one agent is simply "best." Instead, reviewer sentiment is **method-sensitive**, **repository-sensitive**, and connected to PR outcomes through **code-review-specific negativity**. This section should be used as the reviewer-response layer of the team's broader analysis of merge outcomes, review duration, task type, PR complexity, repository context, and developer experience.

Notes for Combining With Team Report

Use the sentiment findings in three places:

Team report area	How to use this section
Results	Include the agent-level sentiment rankings and the method-sensitivity table
Discussion	Explain that sentiment is not a simple leaderboard because general tone and code-review negativity differ

Validity	Mention dataset imbalance, smaller samples for Claude Code/Cursor, and repository sensitivity
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Recommended figures:

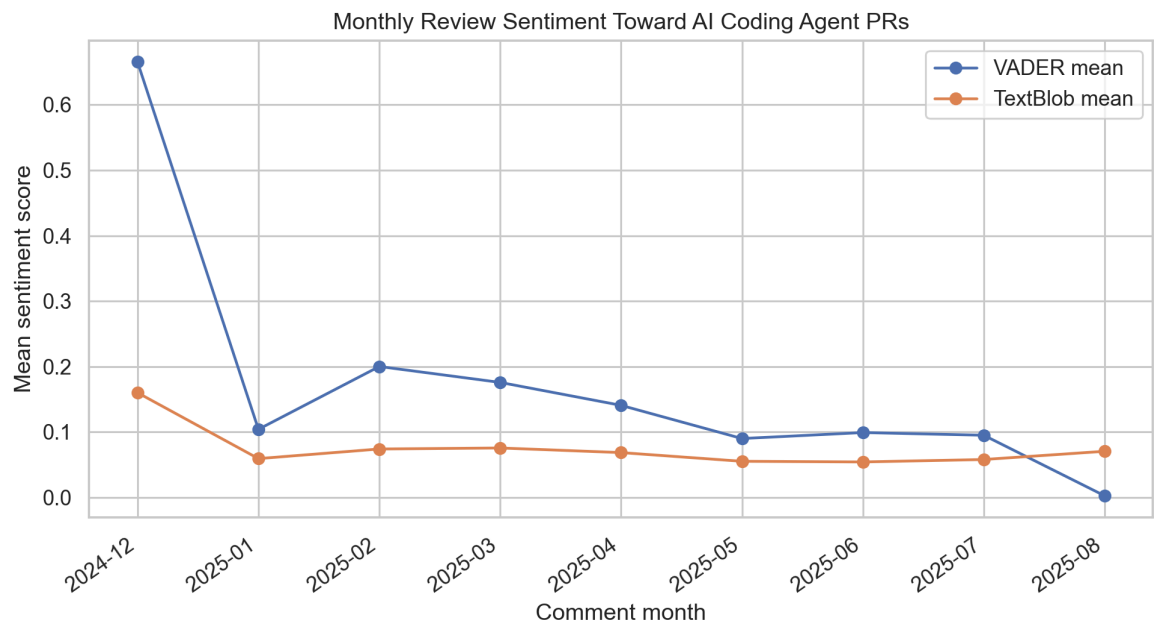
- Monthly sentiment trend
- Method ranking comparison

Short Limitations

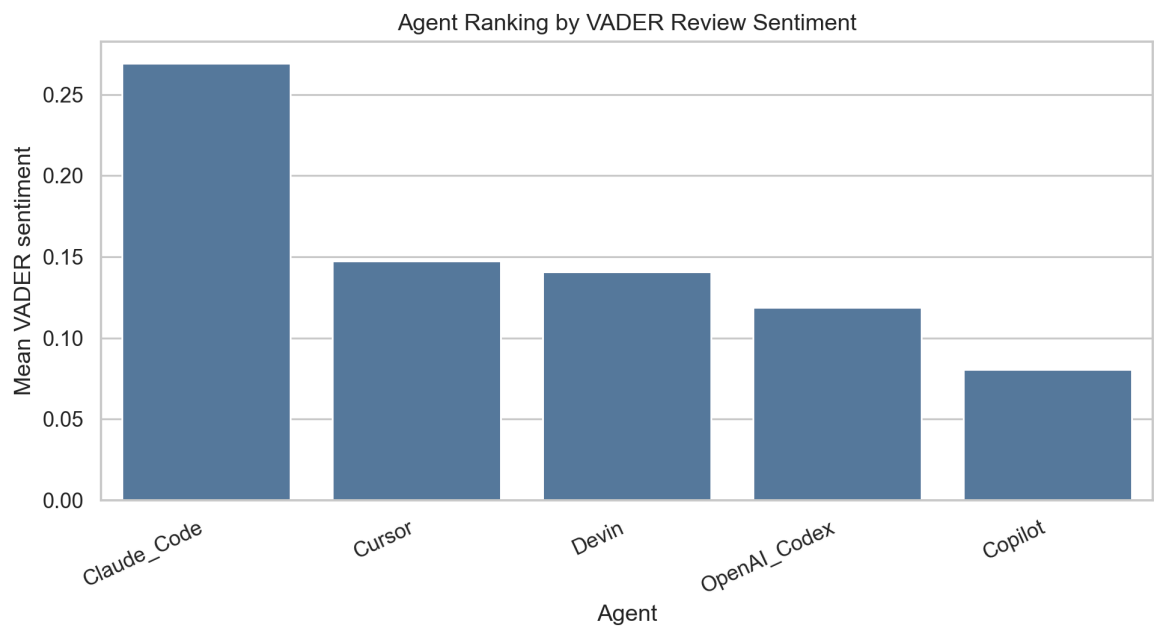
- Sentiment is used as a proxy for reviewer reception, not as a complete measure of trust or code quality.
- General-purpose sentiment tools can misread technical review language, so results are compared with a code-review-specific method.
- The dataset is imbalanced across agents and repositories; therefore, raw rankings are interpreted alongside downsampling and repository-normalized checks.
- Claude Code and Cursor have smaller samples than Copilot and Devin, so their rankings should be interpreted more cautiously.
- Sentiment findings should be interpreted together with merge status, review duration, task category, PR complexity, repository context, and developer experience.

Appendix: Generated Figures

Monthly sentiment trend



Agent VADER sentiment ranking



Ranking comparison across sentiment methods

